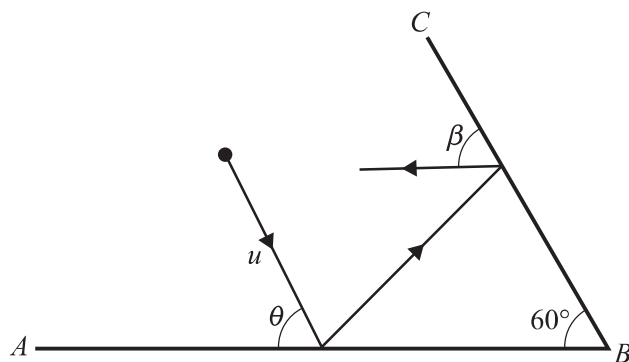




AB and BC are two fixed smooth vertical barriers on a smooth horizontal surface, with angle $ABC = 60^\circ$. A particle of mass m is moving with speed u on the surface. The particle strikes AB at an angle θ with AB . It then strikes BC and rebounds at an angle β with BC (see diagram). The coefficient of restitution between the particle and each barrier is e and $\tan \theta = 2$.

The kinetic energy of the particle after the first collision is 40% of its kinetic energy before the first collision.

- (a) Find the value of e . [4]

This image shows a full page of a document template designed for handwritten notes or essays. It features approximately 28 evenly spaced, thin grey horizontal lines across the entire width of the page. The margins are consistent on all sides, providing ample space for writing. There are no pre-printed questions, headings, or other markings on the page.

[Turn over